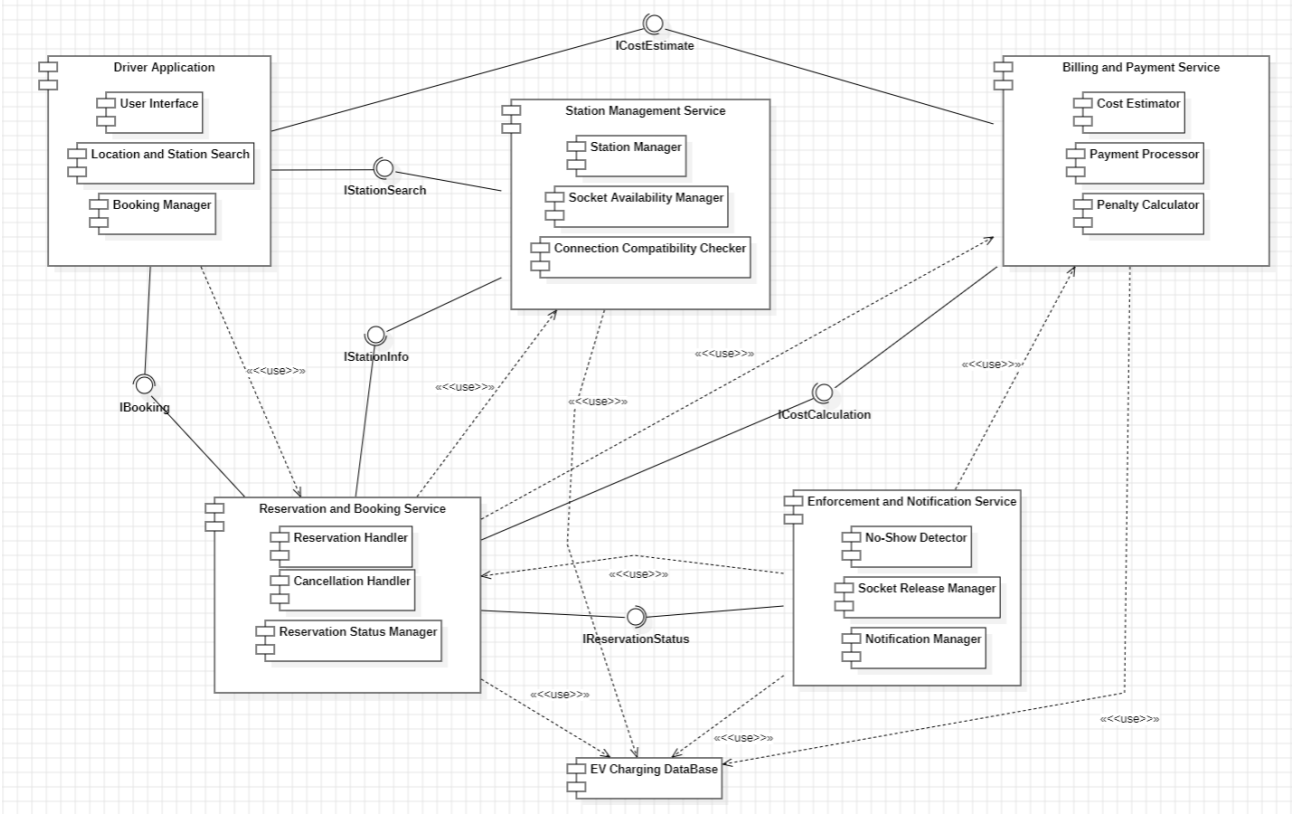
SOFTWARE ENGINEERING LAB 3: Component Modelling & Architectural Pattern Selection

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Case Study Title: EV Charging Station Reservation System



UML Component Diagram of the EV Charging Station Reservation System

Architecture Selection: We chose Microservices Architecture for the EV Charging Station Reservation System.

Architectural Choice: The system is divided into separate services for station management, reservation and booking, billing and payment, and enforcement and notification. Each service handles a specific part of the system.

Reasons:

1) The system has different functions such as finding charging stations, checking socket availability, making reservations, calculating costs, and handling no show cases. Dividing these functions into separate services makes the system easier to manage and update.

2) The system must support at least 500 concurrent reservation requests with a response time of less than 3 seconds. The reservation service can be scaled separately when there are many booking requests without having to scale the whole system.

Security Advantage: Each service can have controlled access to the data and functions it needs. For example, the billing service can be restricted to payment and cost related information. This reduces unnecessary access to sensitive data.

Performance Benefit: Services that receive more requests can be scaled separately. This helps the reservation service handle a high number of booking requests without affecting other services.